



JACQUELINE NYINAWABAGESERA

SOFTWARE ENGINEER

Contact

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Education

AkiraChix
codeHive - Diploma in
Information Technology
February 2025 - Present

Skills

- **Languages:** Python, SQL, JavaScript
- **Libraries & Frameworks:** Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, HuggingFace Transformers, OpenCV, Matplotlib, Seaborn, Plotly, FastAPI, Django
- **Databases & Vector Search:** PostgreSQL, ChromaDB, Supabase
- **Platforms & Tools:** Jupyter Notebook/Lab, Google Colab, Google Cloud, GitHub, Postman, Cloud Deployment

References

Linda Kamau
Founder and Executive Director
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John Owuor
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Personal Statement

The evolution of education throughout the COVID-19 pandemic highlighted the vital role that technology plays in ensuring continuous and inclusive learning. With significant usage of low-tech and no-tech solutions, Jacqueline believes technology holds the answers to the big question; how can the education sector in Africa best prepare for the next pandemic?

Projects

Recos AI

Recos is an AI-powered agent designed to assist recruiters in identifying the ideal candidates through data-driven insights that reduce skill mismatches and enhance hiring accuracy.

Approach and Key Learnings:

- Integrated Odoo recruitment data by building a continuous data synchronization system that enables access to candidate data, job postings, and all related application information stored in Odoo.
- Used AssemblyAI for live transcription in a Google Meet addon with interview analytics, then switched to Google Cloud Speech-to-Text for better accuracy and real-time performance.
- Developed an NLP system using Gemini LLM for skill extraction and semantic matching from resumes, for more reliable and context-aware results.
- Created an interview analysis workflow that performs sentiment analysis, generates follow-up questions, and provides candidate performance reports with hiring recommendations.

Housing Prices Prediction

Developed a regression model to predict housing prices from 2006 to 2010 by preparing and examining data from the Ames Housing dataset from Kaggle.

Approach and Key Learnings:

- Cleaned and preprocessed housing datasets using Pandas, addressing missing values with median and mode imputations to ensure accuracy, reliability, and bias prevention.
- Conducted exploratory data analysis (EDA) with Pandas, Matplotlib, and Seaborn, identifying key correlations such as square footage, overall quality, and neighborhood that strongly influenced house price predictions.
- Trained and evaluated multiple regression models such as Linear Regression, Random Forest, Gradient Boosting, KNN, and SVR using Scikit-learn, evaluating performance with MAE, MSE, RMSE, and R² metrics.

Movie Recommendation System (RAG + AI Agent)

Built a personalized movie recommendation system that suggests films based on user queries by leveraging semantic search and AI-generated explanations to enhance relevance and user engagement.

Approach and Key Learnings:

- Leveraged ChromaDB to embed and index movie metadata, enabling fast and accurate semantic search for content retrieval.
- Used a modular recommendation workflow using Google ADK, facilitating flexible and scalable reasoning through composable agent steps.
- Experimented with different prompting strategies, such as few-shot, zero-shot, and chain-of-thought; identified a hybrid zero-shot and chain-of-thought approach as the most reliable and efficient.
- Gained insights into the critical role of prompt engineering combined with modular agent architecture for building scalable, maintainable recommendation systems.